

Vikash Kumar

AI Automation & Data Operations Engineer

📍 East Delhi, India

📞 +91 92050 89152 | ✉️ vktofly@gmail.com

🌐 linkedin.com/in/vktofly | 🐙 github.com/vktofly | 🌐 vktofly.github.io

SUMMARY

AI Automation & Data Operations Engineer with proven experience architecting scalable data pipelines, autonomous workflows, and backend infrastructure. Specialized in utilizing Python, Docker, n8n, and Supabase to drive operational efficiency. Seeking remote roles in AI Operations, Data Engineering, or MLOps where robust system design, production deployment, and Day-2 observability are critical.

SKILLS

Backend & Automation: Python, Docker, Kubernetes, n8n, Airflow, Supabase, REST APIs, CI/CD

Data & Machine Learning: Agentic AI (LangGraph, CrewAI), RAG, Vector DBs (Qdrant, Pinecone), AI Observability (LangSmith, DeepEval), MLflow, ETL/ELT, SQL

Cloud & Infrastructure: GCP (Google Cloud), Self-hosted infrastructure, Containerization, Git

Core: System Architecture, Production Deployment, Cost & Latency Optimization, Troubleshooting

EXPERIENCE

Mar 2025 – Present

Remote

Founder & Principal Engineer | *Mono Pvt. Ltd.*

- Architected self-hosted agentic workflows (LangChain, n8n, Python, Supabase) and deployed scalable, production-grade LLM solutions, reducing manual processing time by 34%.

Mar 2024 – Jan 2025

Noida, India

AI Software Engineer | *Techype Pvt. Ltd.*

- Deployed and instrumented a scalable talent acquisition platform via AI-assisted development (Cursor, Claude), streamlining candidate sourcing to reduce time-to-hire by 40%.
- Engineered full-stack features from ideation to production deployment using Agile methodologies, ensuring high reliability and aligning technical deliverables with business goals.

Jul 2024 – Jan 2025

New Delhi, India

IT Manager & Operations | *Aparc Healthcare*

- Directed IT infrastructure across a multi-department facility, ensuring 99%+ uptime while providing technical leadership and bridging technology gaps.
- Automated and optimized patient data management workflows using internal tools, improving daily operational efficiency.

PROJECTS

Self-Optimizing Agent Pipeline (DSPy + LangGraph) [\[GitHub\]](#)

DSPy, LangGraph, Python

- Architected a cyclic, multi-step RAG agent where prompts are programmatically optimized and compiled by DSPy, eliminating manual prompt engineering and improving accuracy by [X] %.
- Orchestrated complex agentic workflows using LangGraph to achieve highly resilient and deterministic generative AI pipelines, coordinating [X]+ distinct sub-agents.

Embedding Data Drift Monitor [\[GitHub\]](#)

Qdrant, Prefect, Python

- Engineered and deployed an MLOps pipeline using Qdrant and Prefect to track the distribution of [X]M+ vector embeddings over time, computing statistical divergence (MMD & Centroid Shift).
- Automated Slack alerts based on data degradation thresholds, reducing incident response time by [X] % and showcasing a robust approach to Day-2 operations.

High-Throughput Local Inference Engine [\[GitHub\]](#)

FastAPI, Prometheus, LLMs

- Architected a FastAPI-based OpenAI-compatible inference server running locally, integrating Prometheus middleware to track throughput and latency, achieving [X]+ tokens/second.

- Load-tested and optimized continuous batching under high-traffic workloads, proving cost-effective deployment by reducing inference costs by [X]%.

EDUCATION

2020 – 2023	M.Tech in AI/ML (online, self paced) <i>IIT & MIT OpenCourseWare</i>
2016 – 2020	B.Tech in Computer Science and Engineering <i>GB Pant Govt. Engineering College (GGSIPU) – 74%</i>
2012 – 2015	12th & 10th (74%) <i>CBSE Board – Advanced Mathematics and Sciences (PCM)</i>

CERTIFICATIONS & ADDITIONAL INFO

Certifications: Master of Artificial Intelligence (UPenn), Docker for Absolute Beginners (Coursera), GitHub Copilot for Beginners (Coursera), Data Analysis Using PySpark (Coursera), Google Data Analytics Capstone (Google)

Languages: English (Fluent), Hindi (Native)